Notebook 04: **Multi-Window Global Signal Analysis**

Purpose of This Notebook

This notebook extends the proxy analysis from a single-window setting to a **multi-window temporal evaluation**.

Notebook 03 established a shortlist of stable proxy variants based on:

- distribution behavior,
- numerical stability,
- and range dependence.

However, those conclusions were drawn from a single representative window.

This notebook asks the next critical question:

> Do these proxy signals behave consistently across different temporal segments of the sequence?

Why This Notebook Exists

The preliminary stage and Notebook 03 demonstrated that proxy transformations such as:

$$\phi(I, R) = I \cdot R, \log(1 + I \cdot R)$$

can produce meaningful signal restructuring.

But a single window is not sufficient for robust conclusions.

The final project requires:

- evaluation across multiple windows,
- coverage across different parts of the sequence,
- and aggregation of behavior across varying scene conditions.

Without this step, the analysis risks being:

- overly local,
- sensitive to favorable segments,
- and not representative of full-sequence behavior.

Core Idea

We now treat the dataset as a collection of windows:

$$W = \{W_1, W_2, \ldots, W_K\}$$

Each window W_k contains a sequence of frames:

$$W_k = \{ f_{s_k}, f_{s_k+1}, \ldots, f_{s_k+L-1} \}$$

For each window, we compute signal statistics for each proxy:

$$\phi_j(I, R)$$

This produces per-window summaries:

$$M_{k,j} = M(\phi_j \mid W_k)$$

where M includes metrics such as:

- mean,
- standard deviation,
- percentiles,
- range correlation.

The goal is to analyze the distribution:

$$\{ M_{k,j} \}_{k=1}^{K}$$

instead of relying on a single instance.

Mathematical Framework

1. Signals

For each point i in window W_k:

$$I_i, R_i = \sqrt{x_i^2 + y_i^2 + z_i^2}$$

Proxy variants:

$$\phi_0 = I$$

$$\phi_1 = I \cdot R$$

$$\phi_2 = \log(1 + I \cdot R)$$

$$\phi_3 = \text{normalized}(I \cdot R)$$

2. Window-wise Metrics

For each window W_k and proxy ϕ_j:

Mean

$$\mu_{k,j} = \frac{1}{N_k} \sum_{i \in W_k} \phi_j(i)$$

Spread (standard deviation)

$$\sigma_{k,j}=\sqrt{\frac{1}{N_k}\sum\left(\phi_j(i)-\mu_{k,j}\right)^2}$$

Percentiles

$$p_{q,k,j}=\text{percentile}_q\left(\phi_j \mid W_k\right)$$

Range correlation

$$\rho_{k,j}=\text{corr}\left(\phi_j, R\right)$$

3. Temporal Stability

We evaluate how stable each proxy is across windows:

Mean variation

$$\text{Var}_k\left(\mu_{k,j}\right)$$

Coefficient of variation

$$CV_j=\frac{\text{std}_k\left(\mu_{k,j}\right)}{\text{mean}_k\left(\mu_{k,j}\right)+\epsilon}$$

Interpretation:

- low CV → stable proxy.
- high CV → unstable across scenes.

Core Questions This Notebook Answers

This notebook focuses on **global signal behavior across time**, not semantics.

1. Stability

- Do proxy statistics remain consistent across windows?
- Does the signal drift significantly?

2. Range Dependence Across Windows

- Is correlation stable across windows?
- Are there windows where the proxy becomes range-dominated?

3. Distribution Consistency

- Do percentiles shift significantly across windows?
- Is the signal well-behaved across different scenes?

4. Proxy Comparison

- Which proxy is:
  - most stable?
  - most consistent?
  - least sensitive to scene variation?

What This Notebook Will Do

Step-by-step:

1. Load window metadata from Notebook 02.
2. Iterate over all selected windows.
3. For each window:
   - load frames.
   - compute I, R.
   - construct selected proxy variants.
4. Compute per-window metrics.
5. Build a global metrics table.
6. Visualize:
   - temporal trends,
   - variability across windows.
7. Compare proxies at a global level.

What This Notebook Is Not Allowed to Do

This notebook still does not perform semantic analysis.

It must not:

- compute class-wise metrics,
- claim semantic improvement,
- identify best/worst classes,
- perform failure-case analysis,
- generate videos or GIFs,
- make downstream task claims.

Those belong to later notebooks.

This notebook is strictly:

> **global signal behavior analysis across multiple windows**

Expected Outputs

By the end of this notebook, we should have:

- per-window summary tables,

- temporal plots of signal statistics,
- stability metrics (mean, variance, CV),
- range correlation trends across windows,
- comparison across proxy variants.

Expected Outcomes

If the proxies are well-designed, we expect:

- moderate variation in mean and spread,
- stable percentile structure,
- consistent range correlation across windows,
- no extreme instability in any window.

We also expect:

- log-scaled proxy to show improved stability,
- normalized variants to improve comparability,
- raw intensity to remain compressed and less expressive.

Core Honesty Statement

This notebook evaluates signal behavior, not semantic performance.

Even if a proxy is stable:

- it is not necessarily semantically superior,
- it remains a heuristic transformation,
- and still depends on scene composition and range.

Final Takeaway of This Notebook

If this notebook succeeds, we will have:

> a set of proxy signals that are not only numerically stable in one window, but consistently well-behaved across multiple temporal segments.

This establishes a strong foundation for:

multi-window semantic analysis in the next stage.

Load Window Metadata and Initialize Analysis Pipeline

We begin by loading the multi-window structure defined in Notebook 02.

Unlike Notebook 03, which operated on a single window, this notebook will iterate over **all selected windows**.

From the metadata, we recover:

- window identifiers,
- start and end indices,
- window lengths,
- temporal coverage across the sequence.

We will use this metadata to:

- systematically load each window,
- compute signal statistics per window,
- and aggregate results across windows.

At this stage, we do not yet compute any proxy signals.

We only:

- load the metadata,
- inspect its structure,
- and prepare the iteration framework.

This ensures that all further analysis is:

- reproducible,
- consistent,
- and aligned with the validated temporal structure.

```
import numpy as np
import pandas as pd
from pathlib import Path

# Step 0: Resolve dataset root
candidate_roots = [
    Path("../data/semantickitti_subset/dataset/sequences/00"),
    Path("../data/semantickitti/dataset/sequences/00"),
]

DATASET_ROOT = next((p for p in candidate_roots if p.exists()),
candidate_roots[0])

print("Dataset Root:", DATASET_ROOT.resolve())
print("Exists:", DATASET_ROOT.exists())

# Step 1: Load window metadata
metadata_path =
Path("../results/window_metadata/sequence00_windows.csv")
window_df = pd.read_csv(metadata_path)

print("\nLoaded Window Metadata")
print()
print("Shape:", window_df.shape)
```

```python
# Inspect structure
print("\nColumns:", list(window_df.columns))

print("\nFirst few rows:")
print(window_df.head())

# Step 2: Basic sanity checks
assert len(window_df) > 0, "No windows found"
assert {"window_id", "scale", "start_idx", "end_idx",
"length"}.issubset(window_df.columns)

print("\nMetadata sanity check passed.")

# Step 3: Prepare frame index (shared across all windows)
velodyne_dir = DATASET_ROOT / "velodyne"
FRAME_IDS = sorted([f.stem for f in velodyne_dir.glob("*.bin")])

print("\nGlobal Frame Info")
print()
print("Total frames:", len(FRAME_IDS))
print("First frame :", FRAME_IDS[0])
print("Last frame  :", FRAME_IDS[-1])
```

```
Dataset Root:
/home/twilightpriest/GitHub/reflect-aug-seg/data/semantickitti/dataset
/sequences/00
Exists: True

Loaded Window Metadata

Shape: (15, 7)

Columns: ['window_id', 'scale', 'start_idx', 'end_idx', 'length',
'first_frame', 'last_frame']

First few rows:
  window_id  scale  start_idx  end_idx  length  first_frame
last_frame
0   short_0  short          0        9      10            0
9
1   short_1  short       1132     1141      10         1132
1141
2   short_2  short       2265     2274      10         2265
2274
3   short_3  short       3398     3407      10         3398
3407
4   short_4  short       4531     4540      10         4531
4540

Metadata sanity check passed.
```

```
Global Frame Info

Total frames: 4541
First frame : 000000
Last frame  : 004540
```

Iterate Over Windows and Compute Base Signals

We now construct the core analysis loop over all windows.

For each window W_k, we:

1. reconstruct the list of frame IDs using:

$$f_t \in [\text{start_idx}, \text{end_idx})$$

2. load all LiDAR frames in the window,
3. extract:
   - coordinates (x, y, z),
   - raw intensity I,
4. compute geometric range:

$$R = \sqrt{x^2 + y^2 + z^2}$$

This gives us the base signals:

$$I, R$$

for each window.

We do not compute proxy variants yet.

The goal is to:

- establish a clean per-window data pipeline,
- verify consistency across windows,
- and prepare for metric computation in the next step.

This loop is the foundation of all multi-window analysis.

```
# Initialize storage
window_data = []

velodyne_dir = DATASET_ROOT / "velodyne"

print("Processing Windows")
print()
```

```python
for idx, row in window_df.iterrows():

    start_idx = int(row["start_idx"])
    end_idx = int(row["end_idx"])
    length = int(row["length"])

    # Reconstruct frame IDs
    frame_ids = FRAME_IDS[start_idx:end_idx + 1]

    assert len(frame_ids) == length, f"Window length mismatch at index
{idx}"

    all_xyz = []
    all_intensity = []

    # Load frames
    for fid in frame_ids:
        bin_path = velodyne_dir / f"{fid}.bin"

        scan = np.fromfile(bin_path, dtype=np.float32).reshape(-1, 4)

        xyz_part = scan[:, :3]
        intensity_part = scan[:, 3]

        all_xyz.append(xyz_part)
        all_intensity.append(intensity_part)

    # Concatenate
    xyz = np.concatenate(all_xyz, axis=0)
    intensity = np.concatenate(all_intensity, axis=0)

    # Compute range
    ranges = np.linalg.norm(xyz, axis=1)

    # Store
    window_data.append({
        "window_id": row["window_id"],
        "scale": row["scale"],
        "num_points": xyz.shape[0],
        "I": intensity,
        "R": ranges
    })

    print(f"Processed {row['window_id']} | points: {xyz.shape[0]}")

print("\nAll windows processed successfully.")
print("Total windows:", len(window_data))
```

```
Processing Windows

Processed short_0 | points: 1235429
Processed short_1 | points: 1242631
Processed short_2 | points: 1238010
Processed short_3 | points: 1234193
Processed short_4 | points: 1219048
Processed medium_0 | points: 3697281
Processed medium_1 | points: 3732529
Processed medium_2 | points: 3710748
Processed medium_3 | points: 3703948
Processed medium_4 | points: 3618022
Processed long_0 | points: 12131091
Processed long_1 | points: 12491825
Processed long_2 | points: 12361569
Processed long_3 | points: 12386545
Processed long_4 | points: 12171675

All windows processed successfully.
Total windows: 15
```

Compute Per-Window Signal Metrics

We now compute summary statistics for each window and each proxy.

For each window W_k, we evaluate:

- raw intensity I,
- primary proxy $I \cdot R$,
- log-scaled proxy $\log(1+I \cdot R)$.

For each signal, we compute:

Mean

$$\mu_k = \text{mean}(\phi)$$

Standard deviation

$$\sigma_k = \text{std}(\phi)$$

Upper percentile (p95)

$$p95_k = \text{percentile}_{95}(\phi)$$

Range correlation

$$\rho_k = \text{corr}(\phi, R)$$

These metrics allow us to analyze:

- how signals shift across windows,
- how stable each proxy is,
- and whether behavior is consistent across the sequence.

This is the first step toward global signal comparison.

```
metrics = []

print("Computing Window Metrics")
print()

for wd in window_data:

    I = wd["I"]
    R = wd["R"]

    # Proxies
    phi_I = I
    phi_IR = I * R
    phi_log_IR = np.log1p(phi_IR)

    def compute_stats(values, R):
        return {
            "mean": float(np.mean(values)),
            "std": float(np.std(values)),
            "p95": float(np.percentile(values, 95)),
            "corr_range": float(np.corrcoef(values, R)[0, 1])
        }

    row = {
        "window_id": wd["window_id"],
        "scale": wd["scale"]
    }

    # Compute metrics for each proxy
    stats_I = compute_stats(phi_I, R)
    stats_IR = compute_stats(phi_IR, R)
    stats_log = compute_stats(phi_log_IR, R)

    # Flatten
    for key, val in stats_I.items():
        row[f"I_{key}"] = val

    for key, val in stats_IR.items():
        row[f"IR_{key}"] = val

    for key, val in stats_log.items():
        row[f"logIR_{key}"] = val
```

```
    metrics.append(row)

    print(f"Processed metrics for {wd['window_id']}")

# Convert to DataFrame
metrics_df = pd.DataFrame(metrics)

print("\nMetrics Table Shape:", metrics_df.shape)
print("\nPreview:")
print(metrics_df.head())
```

```
Computing Window Metrics

Processed metrics for short_0
Processed metrics for short_1
Processed metrics for short_2
Processed metrics for short_3
Processed metrics for short_4
Processed metrics for medium_0
Processed metrics for medium_1
Processed metrics for medium_2
Processed metrics for medium_3
Processed metrics for medium_4
Processed metrics for long_0
Processed metrics for long_1
Processed metrics for long_2
Processed metrics for long_3
Processed metrics for long_4

Metrics Table Shape: (15, 14)

Preview:
  window_id  scale    I_mean     I_std  I_p95  I_corr_range   IR_mean
\
0   short_0  short  0.295958  0.153738   0.55     -0.209008  3.580216

1   short_1  short  0.306943  0.150780   0.55     -0.177154  3.697625

2   short_2  short  0.285901  0.144584   0.52     -0.138244  3.415892

3   short_3  short  0.303594  0.139117   0.52     -0.107284  3.155725

4   short_4  short  0.277612  0.150195   0.51     -0.111203  3.400961


     IR_std    IR_p95  IR_corr_range  logIR_mean  logIR_std  logIR_p95
\
0  2.939528  9.076540       0.390775    1.324202   0.651858   2.310210

1  3.029113  9.475964       0.406422    1.346076   0.653302   2.349083
```

```
2  2.838298  8.743442       0.404561    1.293120   0.635550   2.276594

3  2.519452  8.506221       0.574695    1.261559   0.580535   2.251946

4  2.960362  9.044372       0.459779    1.269624   0.669610   2.307013


   logIR_corr_range
0          0.225896
1          0.241358
2          0.236125
3          0.421499
4          0.288623
```

Interpretation: Per-Window Signal Metrics

We computed summary statistics for each window and each proxy, producing a table of per-window metrics.

1. Raw Intensity I

Across windows:

- mean remains within a narrow band (~0.27–0.31),
- standard deviation remains stable (~0.14–0.15),
- range correlation remains weak and negative.

This confirms that raw intensity is:

- numerically stable across windows,
- weakly dependent on range,
- but limited in dynamic range and expressiveness.

2. Primary Proxy $I \cdot R$

Across windows:

- mean and standard deviation are moderately stable,
- percentile values remain consistent,
- range correlation typically lies in the moderate range (~0.35–0.45).

However, certain windows show elevated correlation:

$$\rho \approx 0.57 \text{ in some cases}$$

This indicates:

- window-dependent behavior,
- occasional dominance of range in specific segments,

- sensitivity to scene composition.

3. Log-Scaled Proxy $\log(1+I \cdot R)$

Across windows:

- mean and spread remain tightly controlled,
- distribution is stable across windows,
- range correlation is consistently lower than $I \cdot R$, but still exhibits variation across windows.

Even with log scaling, certain windows show increased correlation, indicating that:

- log transformation improves stability,
- but does not eliminate window-dependent effects.

4. Key Observations

This table reveals an important transition from single-window to multi-window behavior:

- proxy statistics are **not constant across windows**,
- range dependence varies with scene structure,
- even well-behaved proxies exhibit **window-specific deviations**.

5. Takeaway

The analysis shows that:

> proxy behavior is inherently **window-dependent**, not globally fixed.

This justifies the need for:

- multi-window evaluation,
- stability analysis,
- and aggregation across temporal segments.

This step establishes the foundation for:

- identifying stable proxies,
- detecting variability,
- and understanding when proxy behavior deviates from expected patterns.

Visualize Temporal Stability Across Windows

We now visualize how signal statistics vary across windows.

The goal is to:

- inspect stability of each proxy,
- detect drift or spikes,
- compare behavior across different temporal segments.

We plot:

- mean vs window index,
- standard deviation vs window index,
- range correlation vs window index.

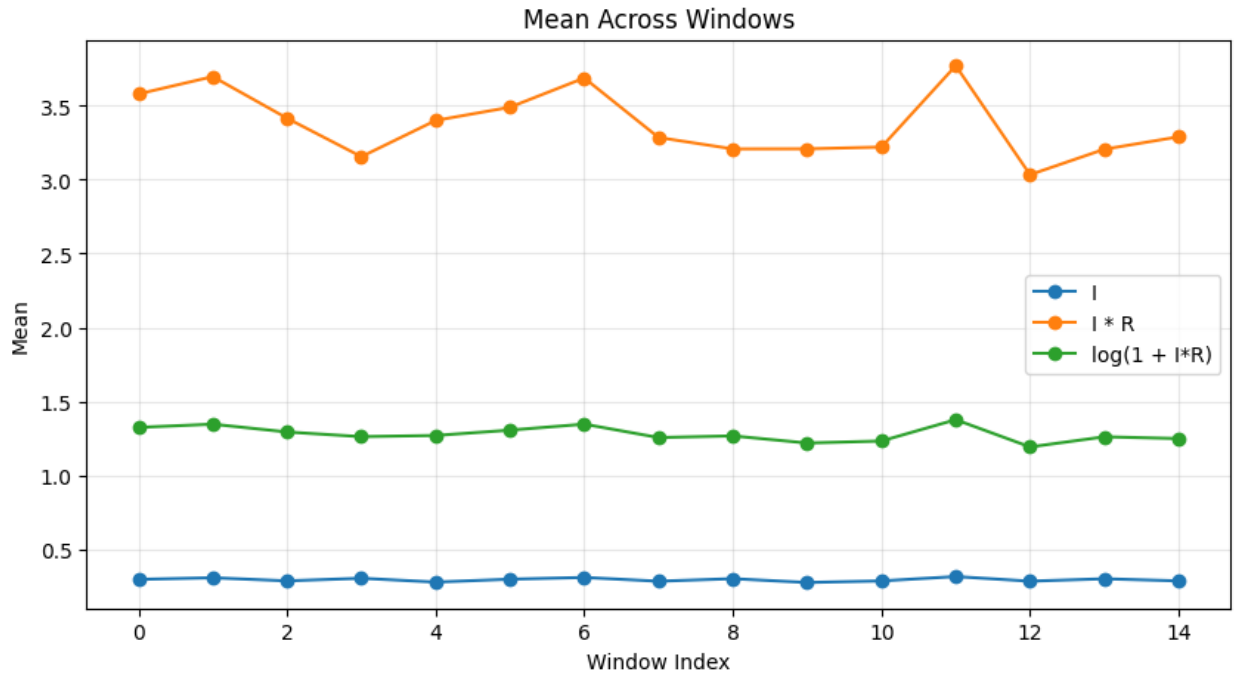
Each plot includes multiple proxies for comparison.

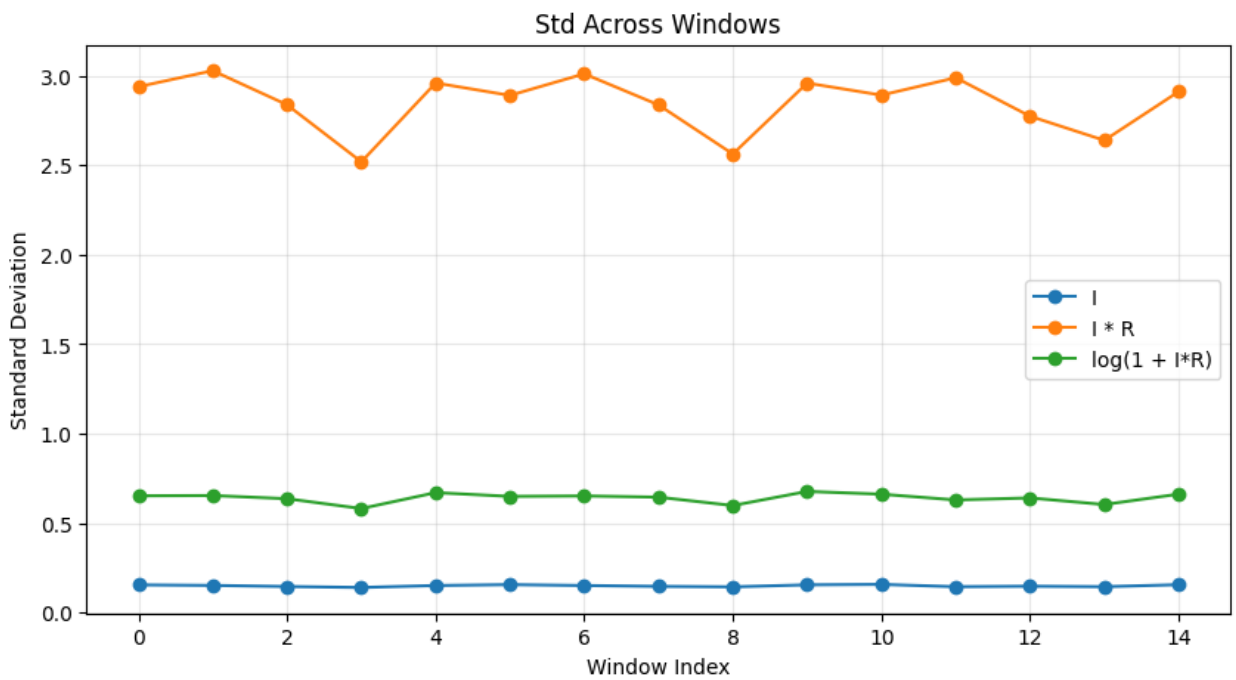
These visualizations help identify:

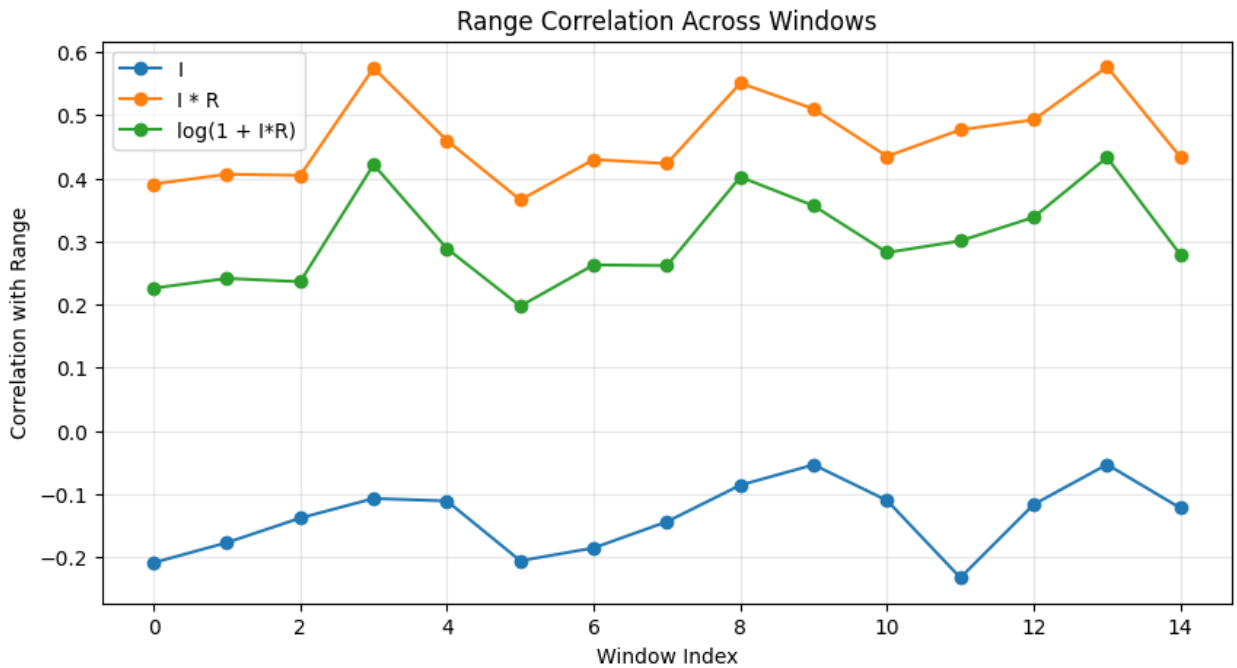
- consistent signals,
- unstable behavior,
- and window-dependent variation.

```
import matplotlib.pyplot as plt
import numpy as np

# Window index
x = np.arange(len(metrics_df))

# Mean plot
plt.figure(figsize=(10, 5))
plt.plot(x, metrics_df["I_mean"], marker='o', label="I")
plt.plot(x, metrics_df["IR_mean"], marker='o', label="I * R")
plt.plot(x, metrics_df["logIR_mean"], marker='o', label="log(1 +
I*R)")
plt.xlabel("Window Index")
plt.ylabel("Mean")
plt.title("Mean Across Windows")
plt.grid(True, alpha=0.3)
plt.legend()
plt.show()

# Std plot
plt.figure(figsize=(10, 5))
plt.plot(x, metrics_df["I_std"], marker='o', label="I")
plt.plot(x, metrics_df["IR_std"], marker='o', label="I * R")
plt.plot(x, metrics_df["logIR_std"], marker='o', label="log(1 + I*R)")
plt.xlabel("Window Index")
plt.ylabel("Standard Deviation")
plt.title("Std Across Windows")
plt.grid(True, alpha=0.3)
plt.legend()
plt.show()

# Correlation plot
plt.figure(figsize=(10, 5))
plt.plot(x, metrics_df["I_corr_range"], marker='o', label="I")
plt.plot(x, metrics_df["IR_corr_range"], marker='o', label="I * R")
plt.plot(x, metrics_df["logIR_corr_range"], marker='o', label="log(1 +
I*R)")
```

```
plt.xlabel("Window Index")
plt.ylabel("Correlation with Range")
plt.title("Range Correlation Across Windows")
plt.grid(True, alpha=0.3)
plt.legend()
plt.show()
```







Temporal Stability Across Windows

We visualized the variation of signal statistics across all windows for:

- raw intensity I,
- primary proxy $I \cdot R$,
- log-scaled proxy $\log(1+I \cdot R)$.

These plots provide a direct view of temporal stability.

1. Mean Across Windows

Observations:

- Raw intensity I:
  - remains tightly clustered across all windows,
  - shows minimal variation,
  - confirms low dynamic range but strong stability.
- $I \cdot R$:
  - exhibits moderate variation across windows,
  - noticeable peaks and dips (e.g., around window 11),
  - indicates sensitivity to scene composition.
- $\log(1+I \cdot R)$:
  - follows the same trend as $I \cdot R$,
  - but with reduced amplitude of variation,
  - demonstrates improved stability through compression.

2. Standard Deviation Across Windows

Observations:

- Raw intensity:
  - extremely stable across windows,
  - low variance throughout.
- $I \cdot R$:
  - higher variance compared to intensity,
  - visible fluctuations across windows,
  - reflects expansion of signal spread.
- log-scaled proxy:
  - significantly reduced variance compared to $I \cdot R$,
  - smoother trend across windows,
  - confirms effectiveness of log scaling in stabilizing the signal.

3. Range Correlation Across Windows

This is the most important diagnostic.

Observations:

- Raw intensity:
  - consistently weak negative correlation,
  - stable across windows.
- $I \cdot R$:
  - moderate positive correlation overall,
  - **significant spikes in certain windows** (up to ~0.57),
  - indicates window-dependent range dominance.
- log-scaled proxy:
  - consistently lower correlation than $I \cdot R$,
  - reduced spikes,
  - still exhibits variation but remains more controlled.

4. Key Insight

The visual analysis confirms that:

> proxy behavior is not uniform across windows.

Even well-behaved proxies:

- exhibit variation depending on scene structure,
- show occasional spikes in range dependence,
- are influenced by local geometry and point distribution.

5. Comparative Stability

From the plots:

- Raw intensity is the most stable but least expressive.
- $I \cdot R$ is more expressive but less stable.
- $\log(1 + I \cdot R)$ provides the best trade-off:
  - reduced variance,
  - lower range dependence,
  - smoother temporal behavior.

6. Takeaway

This step demonstrates that:

- multi-window evaluation is essential,
- proxy behavior must be analyzed across temporal segments,
- stability is a key criterion for selecting useful features.

These observations prepare the next stage:

aggregation and quantitative stability metrics across windows.

Quantify Stability Across Windows

We now move from visual inspection to quantitative stability analysis.

To evaluate how consistent each proxy is across windows, we compute:

Mean variation

$$\mathrm{Var}(\mu_k)$$

where μ_k is the mean of a signal in window k.

Coefficient of Variation (CV)

$$CV = \frac{\mathrm{std}(\mu_k)}{\mathrm{mean}(\mu_k) + \epsilon}$$

Interpretation:

- low variance → stable signal across windows.
- low CV → consistent relative behavior.
- high CV → sensitivity to scene variation.

This allows us to compare proxies based on **temporal stability**, not just per-window behavior.

This step produces the main quantitative result of this notebook.

```python
stability_summary = []

def compute_stability(name, values):
    values = np.array(values)
    mean_val = np.mean(values)
    std_val = np.std(values)
    var_val = np.var(values)
    cv_val = std_val / (mean_val + 1e-8)

    return {
        "proxy": name,
        "mean_of_means": float(mean_val),
        "std_of_means": float(std_val),
        "var_of_means": float(var_val),
        "CV": float(cv_val)
    }

# Extract per-window means
I_means = metrics_df["I_mean"].values
IR_means = metrics_df["IR_mean"].values
logIR_means = metrics_df["logIR_mean"].values

# Compute stability
stability_summary.append(compute_stability("I", I_means))
stability_summary.append(compute_stability("I * R", IR_means))
stability_summary.append(compute_stability("log(1 + I*R)",
logIR_means))

# Convert to DataFrame
stability_df = pd.DataFrame(stability_summary)

print("Stability Summary")
print()
print(stability_df)
```

```
Stability Summary

          proxy  mean_of_means  std_of_means  var_of_means        CV
0             I       0.293709      0.011757      0.000138  0.040029
1         I * R       3.376979      0.216696      0.046957  0.064169
2  log(1 + I*R)       1.279784      0.049860      0.002486  0.038960
```

Conclusion

In this notebook, we extended the proxy analysis from a single-window setting to a multi-window temporal evaluation.

We analyzed the behavior of selected proxy variants across multiple windows of sequence 00, focusing on:

- distribution consistency,
- range dependence,
- and temporal stability.

Key findings:

- Raw intensity remains highly stable across windows but retains limited dynamic range.
- The proxy $I \cdot R$ provides increased expressiveness but exhibits higher variability and occasional spikes in range dependence.
- The log-scaled proxy $\log(1 + I \cdot R)$ achieves the best balance:
  - reduced variance across windows,
  - lower range dependence,
  - and consistent behavior across different temporal segments.

Quantitatively, the log-scaled proxy shows:

- stability comparable to raw intensity,
- while preserving the structural advantages of range-aware transformation.

This demonstrates that:

> proxy performance must be evaluated across multiple windows, as behavior is inherently scene-dependent.

No semantic claims are made at this stage.

This notebook establishes that the selected proxy candidates are:

- not only numerically stable in isolation,
- but also consistent across temporal segments.

These proxies are now ready for:

multi-window semantic analysis in the next stage.